Causal Attribution of Pneumothorax Bias in Medical VLMs

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ABSTRACT

Open-weight medical Vision-Language Models (VLMs) have shown strong performance in radiological understanding, making their interpretability and fairness increasingly important for clinical deployment. Recent studies have increasingly investigated performance disparities across demographic and clinical subgroups to assess model fairness and robustness. However, these analyzes remain largely observational where they quantify disparities across subgroups but cannot determine whether model decisions are causally driven by pathological evidence or by spurious shortcut features. To address this issue, we introduce the Latent Bias Tracking Framework (LBTF) for tracking pneumothorax diagnostic bias to its causal source inside medical VLMs without per-feature annotation. The framework consists of three complementary stages. By causally localizing the internal representations responsible for pneumothorax diagnosis, LBTF provides a causal perspective on diagnostic bias beyond conventional fairness evaluation, advancing trustworthy and interpretable medical VLMs.

Keywords: Bias, Pneumothorax, X-ray, Vision-Language Models

1. INTRODUCTION

Medical VLMs' increasing accessibility has accelerated research towards clinical decision support, particularly for high-risk conditions such as pneumothorax, where accurate diagnosis remains challenging because X-ray images are characterized by low contrast, subtle intensity variations, and overlapping anatomical structures.[1,2] Understanding the reliability of their diagnostic reasoning has become an essential prerequisite for safe clinical deployment. Despite these advances, interpretability alone does not explain diagnostic bias.[3] We define diagnostic bias as a model's systematic reliance on spurious cues, namely factors that are statistically correlated with the diagnostic label but are not indicative of the underlying pathology, thereby influencing diagnosis beyond the genuine pathological evidence. Such cues may arise from latent visual patterns that correlate with disease prevalence without constituting causal evidence of pneumothorax. A central question of this work is whether such behavioral reliance reflects a causal mechanism encoded in the model's latent representations.

To evaluate such behavior, recent studies have predominantly assessed fairness by measuring predictive disparities across predefined demographic and clinical subgroups.[4–6] While effective for identifying whether observable disparities exist, these approaches provide limited insight into the internal mechanisms responsible for biased predictions. More importantly, they rely on the availability of metadata. In practice, however, demographic and acquisition-related information is frequently removed during data de-identification to satisfy privacy requirements, rendering subgroup-based fairness evaluation infeasible for publicly available medical datasets.

To address this limitation, we introduce LBTF, a label-free framework for tracing pneumothorax diagnostic bias in medical VLMs without requiring per-feature annotation. We implement and evaluate LBTF using MedGemma-4B-IT[7] and Qwen3.5-4B.[8] LBTF is instantiated over 19,237 chest radiographs from CANDID-PTX dataset.[9] The proposed framework progresses in three complementary stages. First, the behavioral stage extracts the radiological evidence invoked in the model's free-text reasoning and quantifies the extent to which each feature is relied upon beyond what is supported by the reference diagnosis. Secondly, the representation stage localizes the internal representation of each candidate feature by measuring its decodability throughout the decoder hierarchy. Finally, the causal stage performs controlled interventions on the localized latent representations to quantify their influence on the model's diagnostic decisions.

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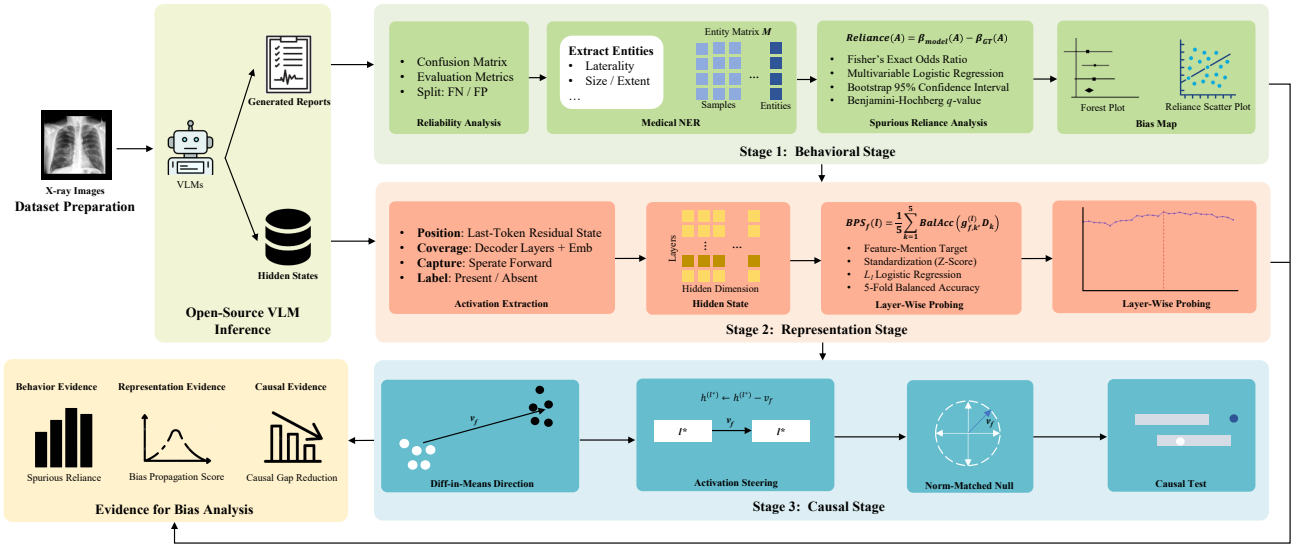


Figure 1. Overview of the proposed LBTF framework.

2. RELATED WORKS

2.1 Mechanistic Interpretability

Recent advances in mechanistic interpretability have shifted the focus toward the analysis of internal representations. Subsequent studies have shown that linear decodability alone does not imply that a representation is functionally used by the model, motivating approaches that distinguish encoded information from causal computation.[10] More recent work has therefore shifted toward feature-level mechanistic interpretability, including sparse autoencoders, dictionary learning, and sparse feature circuit discovery.[11,12] Our work extends this direction by combining representation analysis with causal intervention to distinguish encoded information from representations that influence model decisions.

2.2 Causal Intervention

Causal intervention has recently emerged as a powerful paradigm for understanding the internal behavior. Recently, representation engineering and activation steering have shown that steering hidden states along semantically meaningful directions enables predictable and controllable changes in model outputs, with concept directions estimated using techniques such as contrastive activation vectors, representation engineering, and contrastive activation addition.[13–15] These approaches have established causal intervention as an important tool for mechanistic interpretability in large language models. We build on these intervention techniques to identify and causally validate the latent features responsible for diagnostic bias in medical VLMs.

3. MATERIALS AND METHODS

3.1 Dataset

In this study, our experiments are implemented on the Chest X-ray Anonymized Dataset in Dunedin - Pneumothorax (CANDID-PTX), a dataset for pneumothorax representing 19,237 chest radiographs (3,196 pneumothorax cases) with the resolution of 1024×1024 pixels and age range 16-101 years (mean=60.1 and STDEV=20.1).[9]

3.2 Methods

Figure 1 presents an overview of LBTF framework. The framework proceeds through three stages that escalate from correlation to causation and culminates in three complementary lines of evidence for bias.

3.2.1 Behavioral Stage

The behavioral stage identifies radiological features whose influence on model predictions deviates from underlying pathology. To isolate each feature's independent contribution, we fit multivariable logistic regressions predicting the model output $\hat{y}$ and ground-truth diagnosis y^{GT}:

$$\Pr(\hat{y}=1 \mid \mathbf{x}) = \sigma\Big(\beta_0^{\mathrm{model}} + \sum_{f\in\mathcal{F}} \beta_f^{\mathrm{model}} x_f\Big), \quad \Pr(y^{\mathrm{GT}}=1 \mid \mathbf{x}) = \sigma\Big(\beta_0^{\mathrm{GT}} + \sum_{f\in\mathcal{F}} \beta_f^{\mathrm{GT}} x_f\Big). \tag{1}$$

The spurious reliance score s_f measures the log-odds discrepancy between model reliance and true pathological association:

$$s_f = \beta_f^{\mathrm{model}} - \beta_f^{\mathrm{GT}}. \tag{2}$$

Features with $s_f > 0$ are over-relied upon, while $s_f < 0$ are under-weighted. Confidence intervals and two-sided p-values are estimated via case bootstrapping, with Benjamini–Hochberg q-values controlling the false discovery rate across $\mathcal{F}$.

3.2.2 Representation Stage

To localize where candidate features are encoded for causal intervention, we extract the residual representation $h_i^{(l)} \in \mathbb{R}^d$ $(l = 1, \ldots, L)$ at the final prompt token. For each feature f, we fit an L_1-regularized logistic regression probe on standardized hidden states to predict feature presence $(m_{i,f} = x_{i,f})$. Standardization prevents high-variance dimensions, L_1 regularization promotes feature sparsity, and 5-fold cross-validated balanced accuracy (BalAcc) handles class imbalance.

The layer-wise encoding trajectory is quantified by the Bias Propagation Score (BPS):

$$\mathrm{BPS}_f(l) = \frac{1}{K}\sum_{k=1}^{K} \mathrm{BalAcc}\Big(g_{f,k}^{(l)},\, \mathcal{D}_k\Big), \quad K = 5, \tag{3}$$

where $g_{f,k}^{(l)}$ is the probe trained on layer l of split k, and $\mathcal{D}_k$ is the held-out fold. The layer maximizing $\mathrm{BPS}_f(l)$ is chosen as the target for intervention, with its non-zero probe coefficients identifying the specific latent dimensions encoding feature f.

3.2.3 Causal Stage

To verify whether a localized representation causally drives model decisions, we perform activation steering on false-positive cases $\mathcal{E}_f$. At layer ℓ_f^*, the feature's semantic direction is estimated as:

$$v_f = \mathrm{mean}\Big(h^{(\ell_f^*)} \mid x_f = 1\Big) - \mathrm{mean}\Big(h^{(\ell_f^*)} \mid x_f = 0\Big). \tag{4}$$

Subtracting v_f from the residual stream during autoregressive decoding steers the latent state toward feature absence, yielding modified predictions $\hat{y}_i'$. The raw intervention effect is quantified by the flip rate:

$$\phi_f = \frac{|\{i \in \mathcal{E}_f \mid \hat{y}_i' = 0\}|}{|\mathcal{E}_f|}. \tag{5}$$

To disentangle true semantic influence from non-specific perturbation magnitude, we generate a null flip-rate distribution $\{\phi_f^{(k)}\}_{k=1}^{K}$ using K random directions constrained to $\|r_k\| = \|v_f\|$. The net causal contribution is given by the normalized z-score:

$$z_f = \frac{\phi_f - \mathrm{mean}\Big(\phi_f^{(k)}\Big)}{\mathrm{std}\Big(\phi_f^{(k)}\Big)}, \tag{6}$$

evaluated alongside a permutation test. Additionally, we track the intervention degeneracy rate to ensure that decision flips reflect semantic removal rather than structural decoding failure.

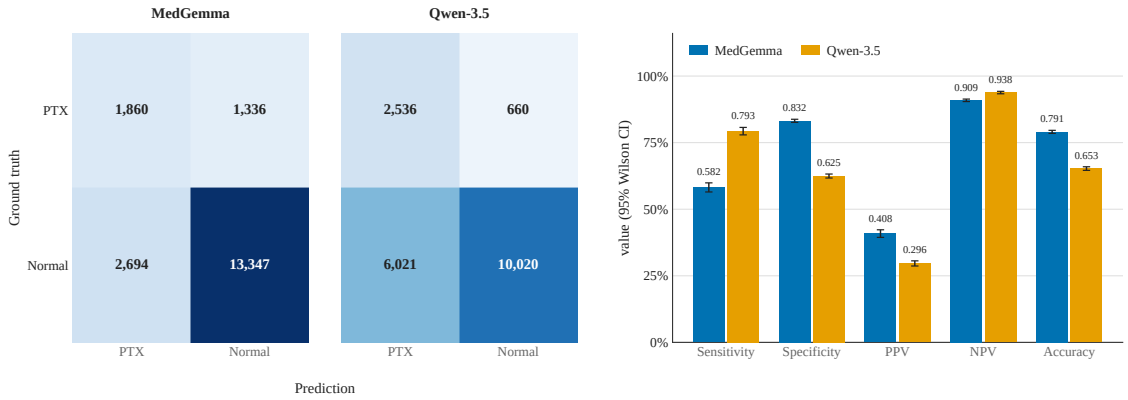


Figure 2. Confusion matrix (left) and metrics (right) of 19,237 pneumothorax diagnosis generated by two models.

4. RESULTS

Diagnosis reliability is presented in Figure 2, where the confusion matrics and metrics for MedGemma and Qwen3.5. Specifically, MedGemma adopts a conservative decision policy that prioritizes high specificity over sensitivity, leading to systematic under-detection of pneumothorax. Conversely, Qwen3.5 operates permissively to maximize sensitivity at the expense of elevated false-positive rates (over-calling). Thus, despite similar aggregate performance, the two models exhibit opposing diagnostic biases.

4.1 Behavioral Shortcut Reliance

MedGemma and Qwen3.5 exhibit different patterns of evidence utilization (Figure 3). MedGemma places disproportionate weight on several entities whose diagnostic relevance is limited, while under-emphasizing multiple findings that are more strongly supported by the ground truth. This imbalance suggests that its predictions are influenced by shortcut cues that receive greater importance than warranted by their clinical utility. In contrast, the entities attracting excessive reliance in Qwen3.5 are largely disjoint from those identified in MedGemma, and several cues emphasized by one model receive comparatively little weight in the other. The divergence therefore reflects not only different degrees of shortcut reliance but also fundamentally different selections of shortcut cues, indicating that shortcut behavior is highly model-specific.

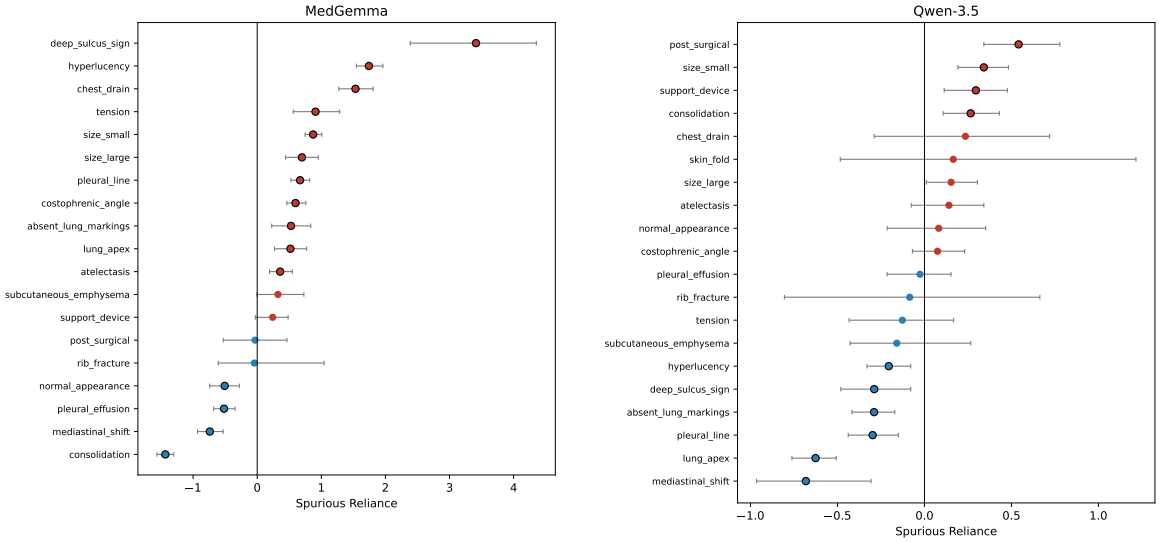


Figure 3. Behavioral bias map showing per-entity spurious reliance for MedGemma (top) and Qwen3.5 (bottom). Red points demonstrate FP-derived driver, blue points demonstrate FN-derived driver and black edge $q < 0.05$.

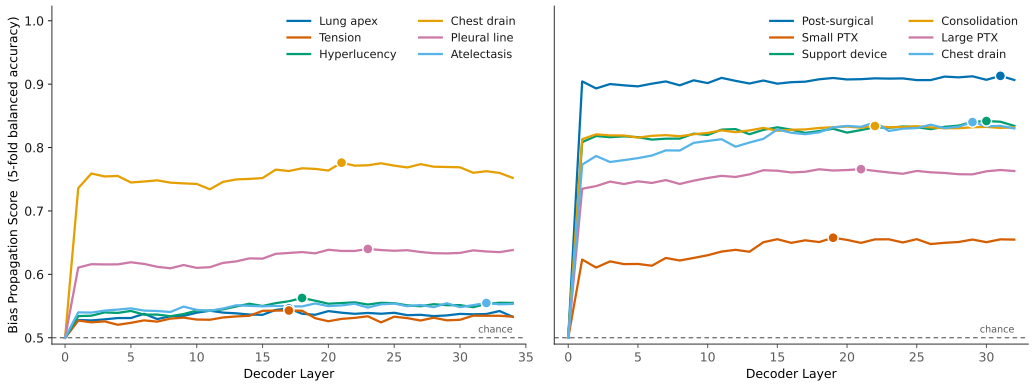


Figure 4. Representation localization in MedGemma (total 34 layers, left) and Qwen3.5 (total 32 layers, right) model.

Feature	s_f	ϕ_{real}	ϕ_{rand}	z_f
MedGemma				
Causally localized ($z_f > 2$)				
Lung apex	0.52	0.350	0.205±.021	6.97
Tension	0.91	0.525	0.421±.043	2.41
Hyperlucency	1.74	0.417	0.330±.041	2.13
Behavioral shortcut, not causal ($z_f < 2$)				
Chest drain	1.53	0.442	0.302±.098	1.42
Pleural line	0.67	0.458	0.435±.026	0.91
Atelectasis	0.36	0.508	0.521±.043	−0.29
Qwen3.5				
Behavioral shortcut, not causal ($z_f < 2$)				
Large PTX	0.15	0.092	0.069±.017	1.29
Small PTX	0.34	0.133	0.110±.022	1.05
Consolidation	0.27	0.108	0.092±.030	0.53
Support device	0.30	0.167	0.161±.028	0.21
Chest drain	0.24	0.059	0.064±.023	−0.21
Post-surgical	0.54	0.192	0.247±.035	−1.56

	MedGemma		**Qwen3.5**	
Feature	ϕ_f	z_f	ϕ_f	z_f
Lung apex	0.350	6.97	0.183	0.99
Tension	0.525	2.41	0.076	−1.14
Hyperlucency	0.417	2.13	0.233	−0.45
Chest drain	0.442	1.42	0.059	−0.21
Pleural line	0.458	0.91	0.192	−3.21
Atelectasis	0.508	−0.29	0.108	0.57

Table 1. Causal attribution for candidate features (left) and same-entity causal comparison across two models (right).

4.2 Representation-Level Localization

Before intervening, shown by Figure 4, we localize each candidate entity within the decoder using the BPS, the layer-wise cross-validated decodability of the entity from the residual stream. The peak layer ℓ_f^* defines the site at which the entity is later steered. Every candidate is decodable well above chance and attains a clear peak in the mid-to-late decoder, confirming that these entities are explicitly represented and providing a principled intervention site. More consequentially, decodability differs markedly between the two models and does not anticipate causal influence.

4.3 Causal Localization

Table 1 presents the causal localization results, showing behavioral reliance does not imply causal influence, and causal drivers are highly model-dependent. For MedGemma, suppressing three features—lung apex, tension, and hyperlucency—significantly reduces false-positive predictions beyond the norm-matched random baseline ($z_f > 2$), localizing its over-diagnosis to specific latent directions. Notably, MedGemma's causal ranking diverges markedly from its behavioral ranking. Conversely, in Qwen3.5, no behaviorally relied feature yields a significant intervention effect, with all responses remaining near the random baseline ($z_f < 2$). This indicates that Qwen3.5's

over-diagnosis stems from a distributed, intervention-resistant decision mechanism rather than localized feature shortcuts.

5. CONCLUSION

In this work, we introduced LBTF, a label-free framework that traces a medical VLM's pneumothorax diagnosis from its observable behavior to its causal source within the network. By combining behavioral analysis of spurious reliance, representation-level localization of encoded cues, and causal verification through norm-matched activation steering, LBTF advances bias auditing beyond identifying where disparities are observed to explaining which internal representations actually drive model decisions. Its central insight is that a cue the model reports, or that a probe can decode, is not necessarily one it causally relies on, making causal verification essential rather than assumed. Requiring no per-feature annotation and deriving all signals from model outputs and withheld ground truth, LBTF provides a reusable and interpretable framework for the trustworthy evaluation of open medical VLMs.

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